EV Market Performance and Insights

Project Report: Data Analytics with Tableau

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1. Introduction

1.1 Project Overview

The electric vehicle (EV) industry is one of the fastest-growing sectors of the global automotive market, driven by environmental awareness, government incentives and advances in battery technology. This project analyses EV market data (sales, pricing, battery specifications, driving range, manufacturing country, customer ratings and CO2 savings) and presents it through two interactive Tableau dashboards and a data story.

- **Dashboard 1 (Market Performance):** KPI cards (total EV sales 1.55B, average price $64,921, average range 324.54 km, 738K tons CO2 saved per year), production output by country and vehicle category, top manufacturers by customer review and sales units, driving range distribution, and 3-year retained value.
- **Dashboard 2 (Technology and Impact):** Sales trend over model years, battery technology performance, charging vs efficiency, and regional environmental impact with CO2 saved.

Tableau Public Link:

https://public.tableau.com/app/profile/hardik.parihar/viz/tableau2_17931236519670/emily?publish=yes

1.2 Purpose

The purpose is to turn raw EV market data into clear, interactive insights that help manufacturers, consultants and researchers understand which vehicle categories drive sales, which battery technologies dominate, how range and charging compare, and where the environmental benefit is greatest, so they can make informed strategic decisions.

2. Ideation Phase

2.1 Problem Statement

EV manufacturers and analysts generate large volumes of sales, battery, manufacturing and customer data, but without a consolidated view it is hard to monitor KPIs such as annual sales, battery capacity, driving range, customer satisfaction, manufacturing distribution and market share. Three stakeholders need tailored answers:

Persona	Role	Need
Emily Johnson	EV Market Strategy Director	Top categories and manufacturers, customer satisfaction, battery adoption, range comparison and high-growth segments for portfolio planning

David Chen	Automotive BI Manager	Manufacturer market share, sales, battery distribution, manufacturing country analysis and ratings to guide clients entering the EV market
Dr. Sarah Williams	Sustainable Mobility Research Analyst	Battery comparison, range improvements, global manufacturing distribution and satisfaction to support evidence-based policy

2.2 Empathy Map Canvas

Quadrant	Emily (Strategy Director)	David (BI Manager)	Dr. Williams (Researcher)
Says	"Which segments should we invest in?"	"Who holds the competitive edge?"	"Which technology drives adoption?"
Thinks	Needs executive-level, fast insights	Clients want proof, not opinions	Policy needs credible evidence
Does	Reviews category sales and satisfaction	Benchmarks manufacturers and regions	Compares batteries, range and CO2 impact
Feels	Pressure to optimise the portfolio	Needs confidence in recommendations	Motivated by sustainability goals

2.3 Brainstorming

Ideas considered and grouped by theme:

- **Market:** Sales by category, trend over years, market share by manufacturer.
- **Product:** Driving range by category, 3-year retained value, price KPIs.
- **Technology:** Battery type vs sales and cycle life, charging power vs efficiency.
- **Sustainability:** CO2 saved by region and drivetrain, production by country.
- **Selected:** Two dashboards (Market Performance, Technology and Impact) joined by a Tableau story.

3. Requirement Analysis

3.1 Customer Journey Map

Stage	User action	Dashboard support
Awareness	Opens the dashboard to see the market at a glance	KPI cards on Dashboard 1
Exploration	Filters by vehicle category and compares charts	Category colour legend and filters
Analysis	Drills into manufacturers, range, battery and charging	Bubble, box, bar and scatter charts

Decision	Picks segments, technologies and regions to focus on	Retained value, battery and CO2 views
Sharing	Presents findings to stakeholders	Tableau Public link and story

3.2 Solution Requirement

Functional Requirements

- Display headline KPIs: total sales, average price, average range, CO2 saved.
- Compare sales, production and retained value across vehicle categories and countries.
- Show manufacturer performance by customer review and sales units.
- Compare battery technologies by sales units and cycle life.
- Show charging power vs efficiency and regional CO2 savings.
- Provide interactive filtering by vehicle category.

Non-functional Requirements

- Clear, readable layout with consistent colour for categories.
- Responsive interaction and fast loading.
- Publicly shareable through Tableau Public.

3.3 Data Flow Diagram

EV dataset (CSV/Excel) → Tableau data source connection → Data cleaning and calculated fields (sales, retained value, CO2 saved) → Worksheets (charts) → Dashboard 1 and Dashboard 2 → Story → Tableau Public.

3.4 Technology Stack

Layer	**Tool**
Data source	EV market dataset (CSV/Excel)
Visualisation	Tableau Desktop
Publishing	Tableau Public
Documentation	Microsoft Word / Claude Docs

4. Project Design

4.1 Problem Solution Fit

The problem is scattered EV data with no single view for decision makers. The solution is a pair of focused interactive dashboards that answer each persona's questions directly, so insights that would take hours in raw tables are available on one screen.

4.2 Proposed Solution

- **Dashboard 1:** Market KPIs, production by country and category, manufacturer reviews vs sales, driving range distribution, retained value.
- **Dashboard 2:** Sales trend, battery technology performance, charging vs efficiency, regional CO2 impact map.
- **Story:** A guided sequence of both dashboards with key observations.
- **Delivery:** Published on Tableau Public for easy sharing.

4.3 Solution Architecture

Data layer (EV dataset) → Tableau data model with calculated fields → Worksheet layer (15 category-coloured charts) → Dashboard layer (Dashboard 1 and Dashboard 2) → Story layer → Tableau Public for end users (Emily, David, Dr. Williams).

5. Project Planning & Scheduling

5.1 Project Planning

Phase	Activities	Output
1. Understanding	Study problem statement and personas	Business questions
2. Data preparation	Load dataset, clean fields, create calculated fields	Tableau data source
3. Visualisation	Build worksheets for each business question	Worksheets
4. Dashboards	Combine worksheets into two dashboards	Dashboard 1 and 2
5. Story and publishing	Build story, publish to Tableau Public	Public link
6. Documentation	Complete report and templates	Final report

6. Functional and Performance Testing

6.1 Performance Testing

Test	Expected result	Status
KPI cards display correct totals	Sales 1.55B, price $64,921, range 324.54 km, CO2 738K tons	Pass
Category colour legend filter	Charts update for selected categories	Pass
Tooltips on charts	Values shown on hover	Pass
Dashboard load on Tableau Public	Both dashboards render fully	Pass
Navigation between dashboards	Tabs switch without error	Pass

7. Results

7.1 Output Screenshots

Insert the Dashboard 1 and Dashboard 2 screenshots here.

Key Findings

- Total EV sales reach 1.55B units at an average price of $64,921, with an average range of 324.54 km and 738K tons of CO2 saved per year.
- India, China and Vietnam lead production output.
- Electric Bus has the highest 3-year retained value (14.5M), followed by Electric Truck, Sports Car (8.69M) and Luxury SUV (6.15M).
- Luxury Sedans and SUVs offer the longest driving range; micro city cars and two/three-wheelers the shortest.
- NMC and LFP batteries lead in sales units; Solid-State and LFP show the longest cycle life.
- Higher-capacity vehicles charge fastest (about 350 kW) but use more energy per km.
- Annual sales are stable from 2018 to 2026, led by Two-Wheeler Scooters.

8. Advantages & Disadvantages

Advantages

- Single interactive view of sales, technology, range and environmental impact.
- Serves several personas with one project.
- Easy to share and explore through Tableau Public.
- Supports data-driven product and investment decisions.

Disadvantages

- Depends on the quality and coverage of the dataset.
- Tableau Public data is publicly visible.
- Some charts are dense with 15 vehicle categories.

9. Conclusion

The project shows how Tableau can convert EV market data into actionable insight. The dashboards reveal which categories and manufacturers lead, which battery technologies dominate, how range and charging trade off, and where CO2 savings are greatest. Together they give strategy, consulting and research users a clear basis for planning, positioning and policy.

10. Future Scope

- Connect live or regularly refreshed data sources.
- Add forecasting of sales and battery adoption.

- Include charging infrastructure and government incentive data.
- Add manufacturer-level drill-downs and market share views.
- Build mobile-optimised dashboard layouts.

11. Appendix

Source Code (if any): Not applicable (Tableau workbook).

Dataset Link: used local .csv dataset

GitHub & Project Demo Link:

- **Demo: https://public.tableau.com/app/profile/hardik.parihar/viz/tableau2_17931236519670/emily?publish=yes**
- **Github : https://github.com/hardikparihar054-cmd/Electric-Vehicle-Market-Performance-Insights**